

# Tan Dai Ngo

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## EDUCATION

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**University of Victoria** — Master of Engineering, Applied Data Science (Co-op) *Victoria, BC | Sep 2025 - Dec 2027*

- **Coursework:** Algorithms and Data Models, Systems for Massive Datasets. **GPA:** 8.78 / 9.00.

**University of Chicago** — Master of Science, Computer Science *Chicago, IL, USA | Sep 2020 - Mar 2022*

- **Coursework:** Applied Software Engineering, Distributed Systems, Databases, Networks, Algorithms.

**University of Washington** — Bachelor of Science, Economics *Seattle, WA, USA | Sep 2015 - Aug 2019*

- **Minors:** Informatics, Applied Mathematics. **GPA:** 3.57 / 4.00.

## EXPERIENCE

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**Software Developer** — T-Mobile (via BeaconFire Inc.) *Bellevue, WA, USA | Jun 2022 - Apr 2024*

- Built and maintained REST endpoints across 5+ RBS (Roaming Business System) microservices in Java and Spring Boot, keeping Cassandra query response times under 3 seconds across multiple data sources.
- Traced 20+ cross-service defects to their root cause and deployed features via Jenkins across 50+ releases.
- Debugged production issues with Splunk and built a Mobileum test-rejection email notifier via MS Graph API.
- Automated reprocessing of rejected roaming tests with weekly retries using Kafka, cutting manual effort by 80%.

**Mathematics and Computer Science Teacher** — APU American International Schools *Da Nang City, Vietnam | Aug - Dec 2024*

- Explained technical material to 50+ students with no prior background, adjusting the pitch in real time.

## PROJECTS

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**Multi-Stage IoT Intrusion Detection** — Python, scikit-learn, XGBoost, PyTorch *Team Capstone, UVic*

- Compared six detection models under a 1% false-positive budget, reaching 99.6% recall at 0.82% FPR.
- Designed a four-condition experiment to separate genuine learning from leakage in a public benchmark.
- Proved the benchmark encodes capture-session identity: the same threshold costs 1% FPR one day, 30%+ another.
- Reported the honest cost rather than the headline: PR-AUC falls from 0.995 to 0.242 on a session-disjoint split.

**Ocean Data Platform with RAG** — Python, Spark, XGBoost, Qdrant, FastAPI *Personal Project*

- Harmonized 10M+ ocean and weather sensor observations from NOAA and ONC into one layered, queryable model.
- Evaluated retrieval over dataset metadata against a held-out query set: hit@k 0.9, term recall 0.85.
- Benchmarked XGBoost against persistence on two same-network signals: real skill on air temperature, none on water.
- Containerized the FastAPI service and vector store with Docker Compose for identical local deployment.

**Distributed Facility Reservation System** — Python, FastAPI, SQLite, pytest *Team Project, UChicago*

- Designed 27 versioned REST endpoints and published OpenAPI documentation for every one of them.
- Wrote 71 pytest tests across the API, database, and reservation-rules layers, run on each change.
- Negotiated one booking contract with four teams, then tested against their independently built services.
- Secured every endpoint behind session tokens checked for freshness and permission scope.

**Anomaly Detection at Scale on Species Data** — Python, PySpark *Team Project, UVic*

- Implemented Isolation Forest from the published algorithm rather than a library, then validated the result.
- Identified a previously undescribed variant of feature-specific swamping and proposed two mitigations.
- Verified linear runtime growth on 1,093,203 records using stratified samples across eight quantile bins.

## TECHNICAL SKILLS

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**Languages:** Python, Java, SQL

**AI Tooling:** Claude Code for test authoring, debugging, refactoring, documentation, and plan validation

**Web and APIs:** REST API design, FastAPI, Spring Boot, OpenAPI documentation, session-based auth

**Data and ML:** pandas, NumPy, scikit-learn, XGBoost, PyTorch, Spark, PySpark, Kafka, Cassandra, Qdrant vector store

**Engineering Practice:** Git, GitHub pull requests, code review, pytest, Docker, Jenkins, Splunk, relational schema design